

**Object Oriented Programming
(CS1004)**

Date: April 6th 2024

Course Instructor

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Sessional-II

Total Time: 1 Hour

Total Marks: 30

Total Questions: 03

Semester: SP-2024

Campus: Lahore

Student Name

Roll No

Section

Signature

CLO2: Design a program using object-oriented programming paradigm.

Question No. 1 (CLO 2, PLO 1, C5)

Marks: 10

Generate the output of the following code

```

class M
{
public:
    virtual void MyMemory() (5) (9)
    {
        cout<<"I forget"<<endl;
    }
    void Disk() (2)
    {
        cout<<"Space"<<endl; (6)
    }
    void Erased()
    {
        cout<<"For good"<<endl; (3) (7) (8)
    }
    void ThisExam()
    {
        Erased();
        MyMemory();
    }
    ~M(){}
};

class N:public M
{
public:
    void MyMemory() (1) (4)
    {
        cout<<"Gone"<<endl;
    }
    void Disk()
    {
        cout<<"Slipped"<<endl;
    }
    virtual void Erased()
    {
        cout<<"Rubbed out"<<endl;
    }
};

void main()
{
    M * m1=new N;
    m1->MyMemory();
    m1->Disk();
    m1->ThisExam();
    cout<<endl;

    M m2=*(new N);
    m2.MyMemory();
    m2.Disk();
    m2.ThisExam();
}
    
```


Question No. 2 (CLO 2, PLO 1, C5)

Marks: 10

Design the solution to the following problem using C++

You are supposed to write the software needed by a company to maintain its computer systems. Each computer in the company has a unique serial_no, it also has a fixed RAM, ROM, processor and manufacturer. A computer can either be a laptop. Each laptop has a screen_size and weight. A computer has many softwares installed on it. Each software has a unique license_no, it also has a cost, is system_software, manufacturer and version.

- Identify the classes needed. [2 marks]
- Identify any inheritance relationships [1 marks]
- Identify any composition relationships [1 marks]
- In which order are the constructors called when a Laptop is initialized [2 marks]
- Give the prototype of the classes such that in an array of pointers of Computers if we call print method laptop info is printed differently if the pointer is referencing a laptop. [4 marks]

Question No. 3 (CLO 2, PLO 1, C5)

Marks: 10 -

Consider the following prototype of course and section classes and write code in C++ to answer the following questions.

```
class Course
```

```
{  
    int courseNumber, creditHours;
```

```
public:
```

```
    void setCourse (int courseNum,int creditHrs);//already implemented
```

```
};
```

```
class Section
```

```
{ int secNumber;
```

```
    Course c;
```

```
public:
```

```
    void setSection (int courseNum,int creditHrs,int section);// to be implemented
```

```
};
```

- What sort of relationship exists between the classes Course and Section? [1 marks]
- Assuming that the setCourse member function of the Course class is already implemented give an implementation of the setSection method of the Section class. [4 marks]
- Write a main program that declares an array of 7 objects of type section and set their values as follows: [5 marks]

Course number: 117 for all sections.

Credit hours: 3 for all sections.

Section number: give each section a unique number from 1-7.